

Simulation of Process Scheduling Algorithms

Operating Systems 1 – Scheduling

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Abstract

In this report, we will present our findings achieved by creating and simulating different scheduling algorithms. We implemented FCFS, RR, SPN, SRT, HRRN and MLFB. The metrics for performance are turnaround time (TAT), normalized TAT and average waiting time. The chosen parameters for MLFB will also be justified.

1 Experimental Setup

This section briefly describes the simulator and the experiments.

1.1 Implemented Scheduling Algorithms

We implemented the following scheduling algorithms:

- First Come, First Serve (FCFS)
- Round Robin (RR) with time slices 2, 4, and 8
- Shortest Process Next (SPN)
- Shortest Remaining Time (SRT)
- Highest Response Ratio Next (HRRN)
- 2 versions of Multi-Level Feedback (MLFB) with 5 levels

1.2 Datasets

We used the provided XML datasets containing processes with:

- process ID,
- arrival time,
- service time.

The experiments were performed on the following datasets:

- Dataset A: 10000 processes
- Dataset B: 20000 processes
- Dataset C: 50000 processes

1.3 Measured Metrics

For each scheduling algorithm and each dataset, we computed:

- average turnaround time (TAT),
- average normalized turnaround time (nTAT),
- average waiting time (T_w).

In addition, for two selected datasets, we generated:

- one $nTAT(T_s)$ graph,
- one $T_w(T_s)$ graph,

where each point represents the arithmetic mean over one percentile of service times.

2 Design and Implementation

2.1 Simulator Structure

The fundamental part of our simulator is a Process, multiple processes are stored in a ProcessList. For reading in the XML input we use a class called ProcessReader, this will return a ProcessList. Then, for each different algorithm, we have a queue and an executor. The executor follows the same 3 steps for each different algorithm. First, it looks for new arrivals, then it selects a next process if necessary and lastly, it executes the current process. The queue stores all the arrived processes and the logic for selecting the next process is also found here. Time progression is very simple, every time the loop gets executed, it represents one jiffy. All statistics are stored in the Process itself, which means the processes are stored in a csv file and used for evaluation later. We used python (matplotlib) to render the graphs.

3 MLFB Configuration

We used an MLFB scheduler with 5 levels. The chosen parameters are:

Parameter	Value / Description
Number of levels	5
Time quantum per level	e.g. 1, 2, 4, 8, 16
New job insertion level	e.g. highest priority queue
Demotion rule	e.g. after full quantum usage
Promotion / aging rule	e.g. none or periodic boost
Priority boost interval	e.g. every 50 jiffies / not used

Table 1: Chosen MLFB parameters.

3.1 Motivation

Explain why these parameters were chosen. For example, discuss the trade-off between:

- responsiveness for short jobs,
- fairness,
- starvation avoidance,
- approximation of SRT-like behaviour.

4 Results

4.1 Global Measurements

Insert the global averages for each algorithm and dataset.

Dataset	Algorithm	Avg. TAT	Avg. nTAT	Avg. T_w
10000	FCFS	540.66	26.94	441.01
10000	RR-2	542.72	5.54	443.07
10000	RR-4	542.64	5.80	442.99
10000	RR-8	542.74	6.40	443.09
10000	SPN	301.58	6.27	201.93
10000	SRT	246.54	1.62	146.90
10000	HRRN	375.74	7.71	276.09
10000	MLFB1	542.91	4.64	443.26
10000	MLFB2	546.33	4.30	446.68
20000	FCFS	536.42	22.97	435.64
20000	RR-2	544.03	5.39	443.25
20000	RR-4	543.93	5.60	443.15
20000	RR-8	543.67	6.06	442.89
20000	SPN	297.24	5.70	196.46
20000	SRT	246.98	1.61	146.20
20000	HRRN	373.03	7.12	272.25
20000	MLFB1	545.42	4.61	444.64
20000	MLFB2	545.94	4.25	445.16
50000	FCFS	505.33	23.26	405.80
50000	RR-2	498.41	5.08	398.89
50000	RR-4	498.50	5.30	398.98
50000	RR-8	498.57	5.76	399.05
50000	SPN	289.58	5.85	190.06
50000	SRT	235.85	1.58	136.32
50000	HRRN	353.07	7.26	253.54
50000	MLFB1	497.94	4.29	398.41
50000	MLFB2	497.62	3.95	398.09

Table 2: Global measurements for all algorithms and datasets.

4.2 Graphs for Selected Datasets

4.2.1 Dataset 10000

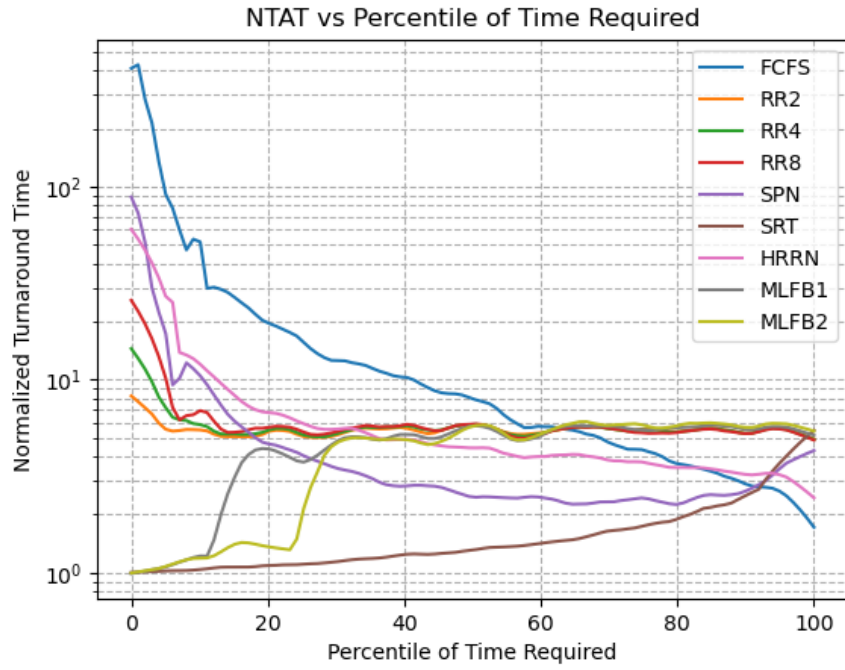


Figure 1: $nTAT(T_s)$ for Dataset 10000.

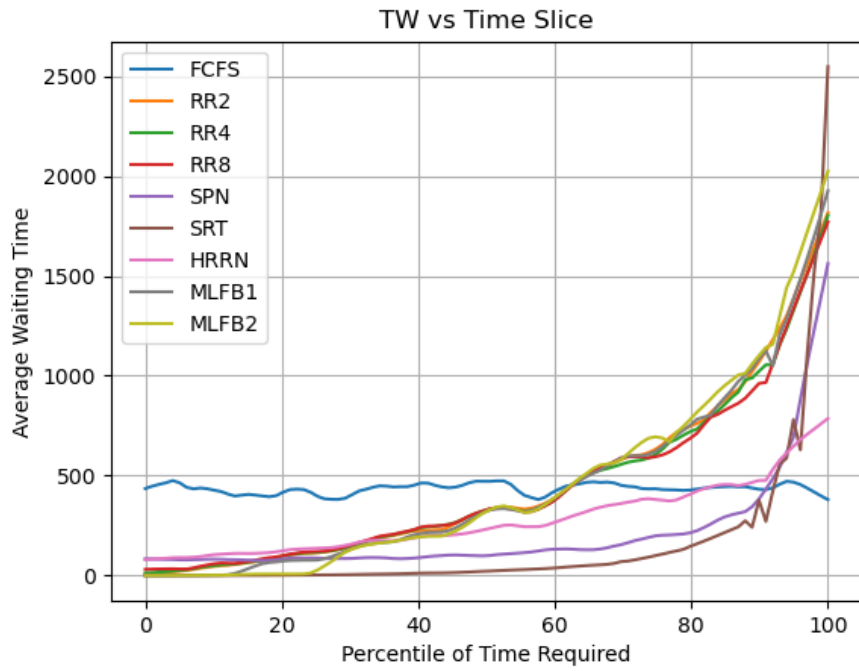


Figure 2: $T_w(T_s)$ for Dataset 10000.

4.2.2 Dataset 50000

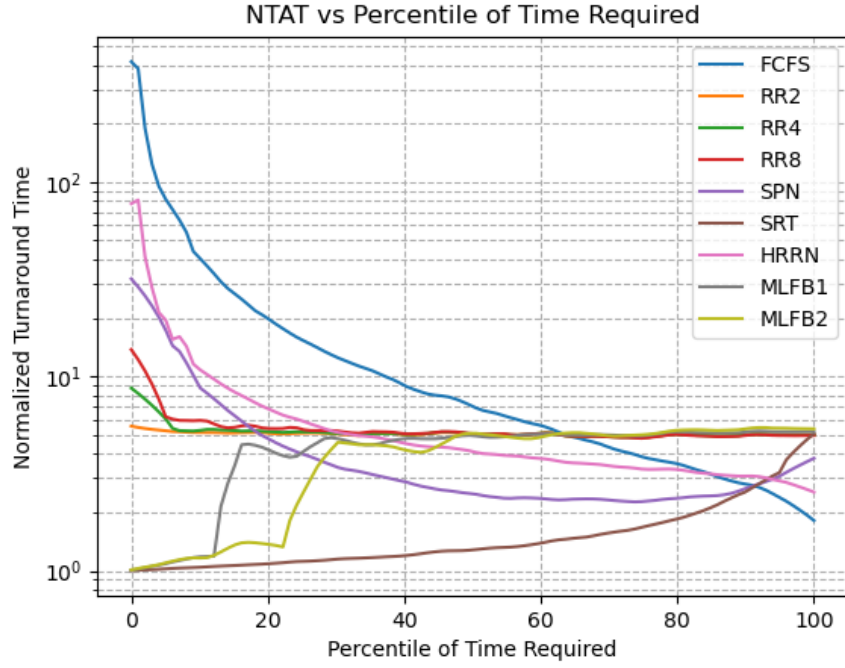


Figure 3: $nTAT(T_s)$ for Dataset 50000.

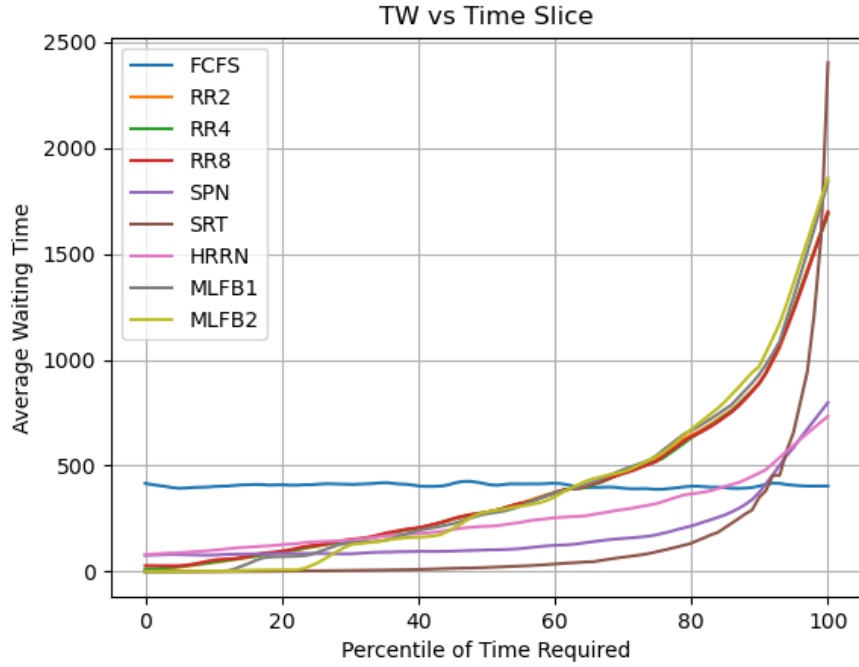


Figure 4: $T_w(T_s)$ for Dataset 50000.

5 Evaluation and Comparison

5.1 Overall Comparison

Discuss the main trends in the global measurements. For example:

- Which algorithms minimize average TAT and nTAT?
- Which algorithms favour short jobs most strongly?
- How does RR change as the quantum increases?
- How competitive is MLFB compared to SPN/SRT?

5.2 Behaviour Across Service-Time Percentiles

Discuss the four graphs:

- performance on short processes,
- performance on long processes,
- fairness,
- whether some algorithms disproportionately penalize certain job sizes.

5.3 Round Robin Analysis

Comment on the effect of time slice choice:

- small quantum $\rightarrow$ more responsiveness,
- large quantum $\rightarrow$ behaviour closer to FCFS,
- trade-off between short-job responsiveness and waiting time for long jobs.

5.4 MLFB Analysis

Evaluate whether your MLFB parameterization achieves its intended goal. Explain:

- which scheduling behaviour it resembles most,
- where it performs well,
- where it performs poorly,
- whether priority boosting or demotion policy had a visible effect.

6 Comparison with Theory

Compare your results with the theoretical behaviour presented in the lectures.

Points to address:

- whether SPN and SRT outperform FCFS and RR for short jobs,
- whether HRRN balances efficiency and fairness as expected,
- whether MLFB approximates short-job preference,
- whether the observed trends match the lecture material qualitatively.

If some results differ from theory, acknowledge this clearly and discuss possible causes, such as:

- implementation errors,
- tie-breaking effects,
- different MLFB parameter settings,
- plotting or averaging choices,
- simulation assumptions.

7 Conclusion

Summarize the most important findings in a few paragraphs. Keep this section focused:

- best-performing algorithms overall,
- trade-offs between fairness and efficiency,
- impact of RR quantum,
- lessons learned from MLFB.